

Wounds Due to Surgery, Injury, or Pressure

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MCG Health
Shared Resources
28th Edition

- Introduction
- Wound Management Tool
 - Wound Management
 - Factors that Influence Wound Healing
 - Debridement
 - Cleansing
 - Dressings
- Types of Wounds
- Nursing Management
- Home and Community Management
- Patient and Caregiver Education
- References
- Footnotes

Introduction

A wound may be described as a loss of continuity of the skin or mucous membrane, and may involve soft tissues, muscle, bone, or other anatomical structures. They are classified by their cause (eg, surgical or nonsurgical; acute or chronic). Wound healing ideally occurs as a uniform process in all connective tissues, and involves the formation of new epithelium and a contraction of underlying healthy granulation tissue to form scar tissue.⁽¹⁾⁽²⁾

This tool contains information about types of wounds and healing. It provides information on how to assess and manage wounds in different levels of care, select dressings, and educate patients and caregivers.

Wound Management Tool

Wound Management

- **Assessment interval**⁽¹⁾⁽³⁾
 - Immediately postoperative
 - Routine shift change evaluation
 - During dressing change
 - Prior to transfer to another facility
 - Admission to post-acute services (eg, recovery facility, home care)
 - At interval in accordance with organization's policy
 - Noticeable wound deterioration (may require more frequent assessment)
 - Development of odor or purulent exudate
 - Other significant change in condition of wound observed
- **Wound type and stage.** See Types of Wounds in this tool for staging and more information.⁽¹⁾
 - Acute
 - Chronic
 - Surgical
 - Nonsurgical
 - Pressure injury - depth of tissue damage and staging
 - Wound closure type⁽¹⁾
 - Primary closure or intention
 - Secondary closure or intention
 - Tertiary closure or intention
- **Wound measurement**⁽³⁾
 - Techniques
 - Ruler method
 - Wound tracing method
 - Digital planimetry method
 - Dimension
 - Length: _____
 - Width: _____
 - Depth: _____
 - Type of exposed skin
 - Nonblanchable redness
 - Partial thickness tissue loss
 - Full-thickness tissue loss with exposure of bone, tendon, or muscle
 - Full-thickness tissue loss without exposure of bone, tendon, or muscle
 - Full tissue loss with base of wound covered by slough or eschar, deepest viable tissue layer unknown
 - Discolored (ie, purple or maroon) intact skin or blood-filled blister

- o Tunneling or undermining
 - Location: _____
 - Depth: _____
 - o Appearance of wound base
 - Granulating or epithelializing
 - Clean
 - Nongranulating
 - Yellow slough or eschar
 - Other: _____
 - o Digital photographs of wounds obtained
- **Complications present**
 - o Wound separation or disruption
 - o Infection noted
 - o Bleeding abnormally
 - o New pressure injury and stage
 - o Causative conditions(3)
 - Impaired nutrition
 - Immobility, especially inability to turn in bed
 - Inadequate circulation
 - Constant moisture presence on skin (eg, urinary or fecal incontinence)
 - Friction and shear on skin
 - Sensory perception - ability to respond to discomfort
 - Other: _____
 - o Complications reported to treating provider
- **Wound treatment**
 - o Pain medication provided pretreatment
 - o Debridement. See Debridement in this tool for more information.
 - Wound was debrided.
 - Wound type
 - Black necrotic wound - presence of necrotic tissue, which may be hardened and is known as eschar
 - Black-yellow wound - some black eschar and some yellow slough
 - Yellow wound - contains necrotic tissue in form of slough
 - Red wound - bright red friable bleeding tissue can be sign of wound infection with invisible bioburden requiring treatment
 - Debridement scheduled
 - Debridement method(4)(5)
 - Autolytic
 - Surgical or sharp dissection
 - Mechanical
 - Enzymatic
 - Biological
 - Other: _____
 - o Cleansing. See Cleansing in this tool for more information.
 - Solutions
 - Saline
 - Other: _____
 - Technique
 - Sterile
 - Clean
 - o Dressings. See Dressings in this tool for more information.
 - o Compression therapy(6)
 - Good skin care at bandage/stocking changes, including washing limb and adding emollients
 - Encourage walking and limb elevation.
 - More adherence with compression stockings
 - o Negative pressure wound therapy (NPWT)(5)(7)
 - Considerations include:
 - Comfort for patient
 - Dressing change every 48 to 72 hours for non-infected wound; more often for infected wound
 - Decreases edema
 - Increases blood supply and oxygenation
 - Application is simple but requires training
 - Use of portable device in patients with restricted mobility (eg, acute and traumatic wounds or chronic open wounds)
 - Used in combination or in sequence with traditional reconstructive procedures
 - May adhere to some wounds
 - Assessment during dressing changes should include observance for arterial bleeding.
 - o Hyperbaric oxygen therapy (HBOT)(2)
 - Adjunctive therapy
 - Uses may include:
 - Acute arterial obstruction post revascularization
 - Complex diabetic foot ulcers
 - Crush injury
 - Failed flap or skin graft
 - Refractory osteomyelitis

- Radiation therapy-induced ulcers or mucosal injury
- Patient and caregiver education completed. See Patient and Caregiver Education in this tool for more information.(8)
- **Impaired healing ability factors** present (check all that apply).(1) See Factors that Influence Wound Healing in this tool for more information.
 - Tissue perfusion, including oxygenation and hydration inadequate
 - Inflammation lasting longer than 5 days
 - Significant contamination present
 - Surrounding tissue status poor
 - Edema
 - Cellulitis
 - Maceration
 - Breakdown
 - Other: _____
 - Undernutrition or malnutrition secondary to(9)(10)(11):
 - Chronic condition (eg, diabetes, malignancy, vascular disease)
 - Cognitive impairment
 - Communication impairment
 - Decreased appetite
 - Decreased thirst response
 - Dependency on help for eating
 - Economic barriers (eg, food insecurity, food desert)
 - Fluid restriction (eg, fear of incontinence, fear of choking)
 - Gastrointestinal losses
 - Imbalanced meals
 - Malabsorption
 - Medications (eg, polypharmacy, side effects, corticosteroid use)
 - Nonadherence with medical regimen
 - Pain
 - Psychological factors (eg, depression, stress, substance abuse)
 - Sensory impairment
 - Compartment syndrome signs(12)
 - Increasing edema
 - Paresthesia
 - Paralysis
 - Diminished peripheral pulses
 - Absent peripheral pulses
 - Pale or discolored skin coloring below wound area
 - Pain disproportionate to injury unrelieved by medication
- **Interventions**
 - Tissue perfusion and oxygenation promotion interventions include(1):
 - Prevent heat loss and shivering of patient.
 - Place socks on feet or use blankets as necessary.
 - Increase mobility or change position frequently.
 - Elevate lower extremities.
 - Medicate for pain.
 - Maintain glucose level control as appropriate for condition.
 - Tobacco use treatment
 - Nutritional interventions include(13)(14):
 - Encourage nutritionally balanced diet.
 - Address impairments to dentition and swallowing.
 - Address physical or cognitive deficits.
 - Provide nutrition as early as possible after surgery or development of wound to meet metabolic demands, prevent or reverse malnutrition, and enhance wound healing.
 - Consider alternative forms of nutrition if unable to tolerate oral diet (eg, enteral feedings, parenteral nutrition).
 - Encourage adequate oral intake in diet that provides adequate protein; calories; vitamins A, C, and B complex; zinc; and iron, where possible.
 - Nutritional status improvement strategies, which may include:
 - Liberalize previous diet restrictions where safe and appropriate.
 - Encourage family and friends to provide favorite foods.
 - Audit and address specific food under-consumption.
 - Encourage intake of 1.25 to 1.5 grams of protein/kg of weight per day.
 - Encourage protein supplements if indicated.(15)
 - Pain management interventions, as appropriate
 - Medication
 - Heat application
 - Cold pack application
 - Other: _____
- **Documentation**
 - Dressing labeled with date and time (if able)
 - Wound assessment data
 - Diagram
 - Measurements
 - Location
 - Photograph

- Drainage characteristics (eg, amount, color)
- Procedure information
 - Date and time
 - Detail of irrigation and dressing
- Patient and caregiver education provided

Factors that Influence Wound Healing

- Factors influencing healing process and likelihood of infection include:
 - General factors
 - Nutrition(5)(13)(16)
 - Acute injury or illness leading to stress response with hypermetabolism and increased catabolism is the leading cause of protein-energy malnutrition, which interferes with adequate wound healing.
 - Even mild nutritional deficiencies may impair healing.
 - Perfusion and oxygenation of tissues
 - Neuroendocrine function
 - Pain and stress
 - Wound healing is affected by pain and stress.
 - Wound pain can be related to exposed nerves, edema, pressure, infection, and injury.
 - Adequate pain management is important to modulate effects of stress.
 - Positioning
 - Nonadherence to care plan
 - Disease process or conditions
 - Smoking
 - Cardiovascular disease
 - Renal failure
 - Immunosuppression
 - Malignancy
 - Remote infection or bacterial infection present
 - Diabetes
 - Systemic steroid use
 - Obesity
 - Age extreme (ie, very young or old)
 - Anemia

Debridement

- Methods of debridement(1)(17)
 - Autolytic - occurs naturally but may be helped by hydrocolloid or hydrogel dressings
 - Surgical or sharp dissection: - excision or wide resection of necrotic tissue, which may necessitate removal of viable tissue. Acute wound has to be trimmed of all questionably viable tissue and foreign material, and chronic wound must be converted to acute wound.
 - Mechanical - physical removal of debris from wound and includes wet-to-dry dressings, wound irrigation, and whirlpool therapy
 - Enzymatic or chemical - application of topical enzymes (eg, collagenase) or chemicals to dissolve necrotic tissue; then covered with moist dressing.
 - Biologic - larval therapy (eg, maggots) to digest necrotic tissue, slough, and bacteria
- Choice of method depends upon:
 - Patient's condition
 - Wound type
 - Amount of tissue to be removed
 - Setting of treatment
 - Patient's ability or willingness to adhere to prescribed treatment plan

Cleansing

- Important factors in cleansing include:
 - Solutions
 - Normal saline is solution of choice.
 - Avoid antimicrobial solutions/antiseptics, which are toxic to tissues and interfere with healing.
 - Types of irrigation
 - Compress or soak with saline-soaked gauze.
 - Pour solution over wound.
 - Push or squeeze piston or bulb syringe filled with saline.
 - Irrigate with enough power to overcome forces holding debris to wound, but not enough to damage tissue.
 - Use appropriate protective precautions for contamination.
 - Topical antibiotics
 - Use for signs of local infection or increased bacterial load.
 - Do not use in presence of systemic infection, such as cellulitis.
 - Do not use for prolonged periods of time.
 - Technique
 - Recommendations include using sterile technique for immunocompromised patient and for procedures that invade blood supply.
 - Assess each patient for suitability of use of aseptic technique.

Dressings

Wound dressings provide the environment for healing and pain management. A moist environment is optimal, especially if the moisture contains the body's endogenous fluids and cells. Choose a dressing by: (1) dry wound -- add moisture; (2) draining wound -- absorb the exudate; (3) necrotic wound -- debride it; (4) infected wound -- treat it with antimicrobial therapy.⁽¹⁸⁾ Wounds vary in the amount of exudate produced and the amount of moisture needed. Select a type of dressing with the appropriate absorptive and hydrating capacity for the particular wound. Dressings also vary in the amount of occlusion they provide. Occlusive dressings reduce pain, provide rapid healing, increase epithelialization and granulation, and reduce friction. However, they may increase maceration around the wound, accumulate exudate, and potentially adhere to the wound. The type of adhesive and conformity of the dressing are also important. The fragility of the patient's skin, location of the wound, and need to avoid contact with the wound margins are considerations in selecting a type of dressing.⁽¹⁾⁽⁴⁾

Table 1 provides information on the type of dressing, indications for use, and characteristics of absorption, hydration, and adhesion.⁽⁴⁾⁽⁵⁾⁽¹⁵⁾⁽¹⁹⁾⁽²⁰⁾

Table 1. Dressing Options, Indications, and Characteristics

Dressing Type	Indications	Cautions or Comments	Absorptive Capacity	Hydrating Capacity	Adhesive Ability
Absorption	Stage 2 to 4 pressure injury with drainage	- Absorbs drainage and deodorizes wound - Needs changing 1-2 times daily	High	Low	Requires additional dressing to secure
Alginate and hydrofiber dressings	- Antimicrobial effect - Moderately to heavily exuding wound; will absorb up to 20 times its own weight - Cavity or flat wounds - Stage 3 to 4 pressure injury - Wound dehiscence - Sinus tracts, tunnels, or cavities	- Trauma-free removal - Requires irrigation with each dressing change - Pack loosely due to "gel" that forms - Not appropriate for abdominal wounds with exposed intestinal mucosa - Not appropriate for dry wound bed - Contraindicated for dry eschar, third-degree burns, surgical implantations, and heavy bleeding	High	Medium	Requires additional dressing to secure
Antimicrobial dressings	- Infected wounds: any type (eg, pressure injuries, venous, arterial, diabetic, or surgical) - Malodorous or colonized wounds - Chronic nonhealing wounds	- Reduces odor - Promotes debridement/autolysis - Used as primary dressing - Longer wear time - Duration of treatment limited - Antimicrobial agent may be deactivated - Contraindicated for certain patient groups, such as those containing iodine or silver (see manufacturer's instructions)	Varies	Low	Varies
Collagen	- Antimicrobial effect - Hard to heal or stalled wounds - Provides moist environment - Can be combined with alginate or silver dressings	- Easy to apply - Contraindicated: - Wounds with large amount of exudate - Third-degree burns - Active vasculitis - Many sizes, shapes and forms available	Medium	None	Requires additional dressing to secure
Composites	- Antimicrobial effect - Contain two or more distinct products manufactured as a single dressing	- Can be used as primary (surface wounds) or secondary dressing - Be familiar with product types included in dressing layers	Varies	Varies	Varies

	- Multiple layers serve several functions and may include: contact layer, absorptive layer, cover layer, and adhesive border with a waterproof outer layer - For use as primary dressing for partial-thickness or shallow full-thickness wounds with minimal exudate	- Dressing changes dependent on combination of products used - Should not be cut - May be used on full-thickness wounds to facilitate autolytic debridement			
Contact layer dressing	- Wound bed protection - For painful wound changes - Nonadherent single layer - Fragile wounds with easy bleeding - Use with secondary absorbent dressing	- Easy to apply - Contraindicated: - Tunneling wounds - Wounds with thick exudate - See manufacturer's instructions for use with third-degree burns	None	None	Requires additional dressing to secure
Foam	- Antimicrobial effect - Exudate management - Around tracheostomy sites - Stage 2 to 4 pressure injuries and dermal ulcers - Tunneling and cavity wounds - Surgical wounds - May be impregnated with charcoal to absorb odor - Provides thermal insulation of wound - May be used with topical medication - Can be used under compression wraps/stockings	- May require secondary dressings or fixation - Does not debride dry necrotic wounds - Not recommended for nondraining wounds or dry eschar - May macerate periwound area - Nonadherent, trauma-free removal/soothing to patient - Easy to apply - Many sizes, shapes and forms available	High	Low	Requires additional dressing to secure
Gauze	- To cover low exudate or superficial wounds - Nonselective debridement, packing or management of	- If removed dry, may disrupt wound healing - May macerate surrounding skin if constantly moist - Additives may cause allergic reaction. - Fibers may shed or adhere to wound bed.	Medium	High, if wet	Requires additional dressing to secure

	- tunnels, tracts, dead space - To keep wound moist when it is frequently inspected - Can be used dry or moist^[A] - Surgical incisions - Burns	- Not recommended for effective moist wound therapy treatment - Many sizes, shapes and forms available			
Hydrocolloids	- Enhances moisture retention - Wound with low-to-medium exudate, or sloughy, cavity, or flat wounds - Stage 1 to 4 pressure injuries and dermal ulcers - To provide protection under Montgomery straps - Can be used as preventive dressing for high-risk friction areas - Can be used under compression wraps/stockings	- Easy to apply - Impermeable to fluids/bacteria - Forms gel when it contacts exudate - Poor oxygen exchange - Extend dressing past wound to obtain adequate adhesion. - Do not use around tracheostomy stomas, tunneling wounds, or wounds with bone or tendons showing. - Can macerate peri-wound area - May leak if exudate heavy; edges roll up - May produce wound odor that can be confused with infection - Contraindicated in third-degree burns - Many sizes, and shapes available	Low	Medium to high	High
Hydrogels	- Enhance hydration - Wounds with low exudate, sloughy/necrotic, malodorous, cavity or flat wounds (eg, stage 2 to 4 pressure injuries, diabetic foot ulcers) - Dermabrasion wound - Painful wounds, radiation burns - To cool wound and to decrease pain	- May be fluid-donating; can macerate peri-wound area - Not recommended for heavily draining wounds - Difficult to apply - Nonadherent, trauma-free removal/soothing to patient - Transparent - Can be used with topical medications - Many sizes, shapes and forms available	Low	Medium to high	Requires additional dressing to secure
Polyurethane film	- To cover low exudate and superficial wounds (eg, blisters, stage 2 pressure injuries, superficial burns) - Donor sites - To secure and protect IV site	- Exudate will pool under dressing, causing excoriation; not for draining wounds. - May strip away new epidermis - Good oxygen exchange - Not for third-degree burns	None	Low to medium	Medium

- To prevent contamination from urine or feces
- To secure gauze dressing
- To decrease friction between skin and bed sheets
- To allow visualization of and protection of stage 2 pressure injuries

• Silver dressings	• Infected wounds • Highly colonized wounds • All wounds except: stage 1 pressure injuries, third-degree burns, and nonexudative wounds • Can be used under compression wraps/stockings	• Inhibits growth of pathogens, especially antibiotic-resistant strains • Must be removed and wound cleansed before MRI • Sensitivity to silver	• High	• Low	• Requires additional dressing to secure
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Types of Wounds

- **Acute** wound examples include:
 - Gunshot wound
 - Stab wound
 - Operative procedure such as surgical wounds, dermatologic wounds, and plastic surgery wounds
- **Chronic** wound examples include(22):
 - Pressure injuries
 - Venous leg ulcers(23)
 - Diabetic foot ulcers
- **Surgical or nonsurgical**
- **Wound classification**
 - Depth of tissue and damage involved. For example, the Wound Ostomy and Continence Nurses Society (WOCN) defines 6 stages of pressure injury.(5)(23)(24)
 - **Stage 1** is characterized by intact skin with nonblanchable redness of a localized area, usually over a bony prominence. May be difficult to detect in individuals with dark skin tones.
 - **Stage 2** is characterized by partial-thickness tissue loss involving dermis. Ulcer is superficial and presents clinically as abrasion, serum-filled blister, or shallow crater.
 - **Stage 3** is characterized by full-thickness tissue loss without exposure of bone, tendon, or muscle.
 - **Stage 4** is characterized by full-thickness tissue loss with exposed bone, tendon, or muscle.
 - Pressure injuries can be difficult to treat and often result in pain, disfigurement, and prolonged hospitalization. Prompt and effective treatment can minimize these deleterious effects and speed recovery.
 - **Unstageable** is characterized by full tissue loss with base of wound covered by slough or eschar, leaving deepest viable tissue layer unknown. Staging may be completed when deepest viable tissue layer is known (eg, after debridement).
 - **Suspected deep tissue injury (DTI)** is characterized by discolored (ie, purple or maroon) intact skin or blood-filled blister from pressure or shear that damaged underlying soft tissue. May be difficult to detect in individuals with dark skin tones.
- **Wound closure**
 - Primary closure or intention
 - Clean wounds with little or no tissue missing, (eg, surgical incision or laceration)
 - Wound edges are approximated manually and held in place by variety of materials (eg, sutures, staples, adhesives).
 - Secondary closure or intention
 - Wounds with irregular edges, those that are contaminated, or both (eg, burns, pressure injuries, traumatic injuries, diabetic ulcers, or chronic wounds such as venous stasis ulcers), and have a large amount of exudate
 - Healing takes place from edges of wound to center and from bottom of wound to more superficial layers.
 - Tertiary closure or intention
 - Wounds intentionally left open, or reopened, due to high bacterial contamination, then closed surgically 4 to 6 days later after bacterial load is reduced
 - Normal healing progresses as secondary closure until it is sutured, then progresses as primary closure.

Nursing Management

- Assessment(19)
 - Provider's order for type and frequency of wound care, including type of dressings and indications for use
 - Surgical wounds may require assessment every 15 minutes during immediate postoperative period and minimum of every shift thereafter.
 - History for allergies, including to latex, tape, or iodine; chronic conditions such as diabetes mellitus; or anticoagulant therapy that might alter way care is given or place patient at risk of wound development
 - Temperature, results of CBC, and wound cultures for indications of infection(25)
 - Skin condition
 - Skin around wound for risk of increase in size or alterations in skin integrity
 - Possible reaction to dressing supplies
 - Pain or discomfort(23)
 - Condition of wound and characteristics, including drainage(1)
 - Measurements (eg, length, width, and depth of wound; periodic photographs of wound)(1)
 - Nutritional status(13)
 - Knowledge of patient and caregivers about condition, understanding of treatment, and prevention
- Treatments and monitoring
 - Wound management
 - Consider need for premedication for pain control.
 - If wound requires dressing(19):
 - Remove old dressing, including old packing, by grasping outer end and gently pulling it from wound, and discarding into biohazard bag (if indicated). Use intermediary to help remove dressing (eg, saline, adhesive remover) to minimize pain and skin trauma.
 - Some wounds with tunneling may have gradual removal of packing over time.
 - Note size and depth of wound and characteristics.
 - Note amount, color, and characteristics of drainage.
 - Cleanse wound using small or large wound techniques. If wound is draining, take care not to contaminate clean areas. Use sterile technique as appropriate.
 - If wound does not require dressing, assess and clean as indicated.
 - If wound requires irrigation(26):
 - Position patient for solution to drain.
 - Provide bedding protection and basin to catch drainage.
 - Protect skin as needed.
 - Using sterile procedure, instill solution into wound.
 - Allow it to drain completely before continuing.
 - When finished, clean and discard all supplies and equipment, placing wet materials into biohazard or waterproof bag.
 - Fragile skin recommendations include:
 - Apply gauze or foam dressing, then wrap with soft gauze.
 - Avoid using tape directly on skin.
 - Educate patient and caregiver as needed.
 - Pressure injury management may require(27):
 - Psychosocial and quality-of-life assessment and management
 - Home assessment(15)
 - Appropriate position, avoiding bony prominences
 - Mobility limitations and needs assessment
 - Assessment of positioning techniques or pressure reduction devices (eg, alternating pressure mattresses)
 - Proper use of equipment or assistive devices
 - Debridement
 - Infection management
 - Nutritional management
 - Monitor for complications, including edema, bleeding, infection, wound dehiscence, or evisceration, and report when indicated.(28)
 - Tetanus booster with traumatic wounds

Home and Community Management

- Discharge planning
 - Discharge requirements(4)
 - Expected outcomes of inpatient care are met.
 - Patient is able to perform self-care needs without assistance or has capable caregiver to help at home.
 - Patient or caregiver is judged to have knowledge and ability to provide care at home.
 - Assessment
 - Assessment of patient and caregiver ability to manage care
 - Be capable of understanding all aspects of therapy and be willing to participate.
 - Have good visual and auditory acuity, good manual dexterity, and strength to perform wound therapy.
 - Learn aseptic technique, including effective hand-washing technique.
 - Assessment of patient's concerns about going home in terms of discharge teaching.
 - Activity
 - Fears
 - Wound pain
 - Wound complications
 - Wound infections
 - Assessment of ability of patient and/or caregiver to provide transportation to physician and clinic appointments
 - Assessment of patient's response to treatment and self-care

- Implementation and coordination of care
 - Treatment plan established by primary care provider or wound care specialist in cooperation with patient and caregiver
 - Clearly defined, measurable goals, and expected outcomes
 - Wound management techniques and procedures, including infection control measures
 - Prevention(15)
 - System for communication, reporting, and follow-up care with primary care provider
 - System for communication, reporting, and follow-up care with wound care specialists (eg, physician, WOCN) and wound care team in wound care clinic
- Consideration of patient's home environment and resources(29)(30)
 - Safe and adequate to support wound care and confirmed prior to hospital discharge
 - In safe area for staff to make home visits
 - Conditions needed for good hygiene
 - Running water, liquid soap, and paper towels are essential for patient to perform proper hand washing before dressing change or wound care treatment.
 - Cleanliness
 - Apply reasonable standard.
 - It is important to have neat and clean work area with separate storage area for equipment and supplies and receptacle for used needles and other sharp objects, as well as soiled dressings.
 - Access to telephone for nursing or emergency assistance if problems occur
 - Proximity to emergency services and support
 - Access to transportation for emergencies or medical follow-up
 - Health status of others living in home
 - Presence of pets and pests that might affect safety of patient in home environment
- Evaluation of care effectiveness
 - Coordinate follow-up provider evaluation (eg, plastic surgeon, WOCN nurse, wound clinic).
 - Wound measurements(1)
 - Wound condition

Patient and Caregiver Education

- Involve patient and caregiver in treatment, prevention strategies, and options.(31)(32)
- Provide preoperative education regarding factors that promote healing of wounds, including:
 - Hand-washing technique
 - Wound splinting when coughing, deep breathing, turning, getting out of bed, or exercising, if indicated
 - Medication management
 - Sterile or clean technique
 - Irrigation or dressing change, if indicated, and how to dispose of contaminated dressings
 - Source for dressing supplies and wound management services
 - Proper nutritional intake(1)(33)
 - Bowel and bladder management, if incontinent
 - Measures for pressure reduction, daily skin examination, and intermittent relief techniques
 - Recognition of signs and symptoms of complications, whom to contact, and how and when to seek medical intervention
 - Redness or red streaks in skin near wound
 - Increase in size of wound
 - Swelling
 - Tenderness
 - Change in color of wound or edges around wound
 - Increased warmth around wound
 - Increased bleeding from wound
 - Foul odor, pus, or unusual discharge
 - Chills or fever
 - Inability to move extremity or joint near or at wound site
 - Nausea and vomiting
 - Diarrhea
 - Inability to eat
- Home management
 - What to look for
 - Healing is from bottom up.
 - New tissue at bottom of wound should be light red or pink and should appear lumpy or glossy.
 - Dead tissue is yellow, black, or tan.
 - There should be less pus as wound heals.
 - Later drainage may be thin, watery, light red, pink, pale yellow, or clear.
 - Keeping wound clean and infection free
 - Effective hand hygiene
 - Methods for cleaning wound
 - Do not clean wound with alcohol, peroxide, iodine, or antibacterial soaps because they slow healing.
 - Keeping wound moist
 - Wound assessment with each dressing change
 - Bandage or dressing to be used
 - Dressing change frequency
 - Keeping wound free of injury
 - Reinforcing preoperative training and return demonstration of wound care

- Health promotion and prevention(1)
 - Importance of adherence to treatment regimen
 - Stress management
 - Lower extremity arterial disease(34)
 - Measures for pressure reduction, daily skin examination, and intermittent relief techniques
 - Stop smoking.
 - Keep legs in neutral or dependent position.
 - Avoid injury resulting from hot soaks, heating pads, medicated corn pads, moisture between toes, friction, tight clothes, going barefoot, or crossing legs.
 - Wear socks and hose and shoes that fit well.
 - Do not wear high-heeled shoes.
 - Do not go barefoot.
 - Exercise ankles and calves often throughout day.
 - Obtain nail care by podiatrist.
 - Lower extremity venous disease(35)(36)
 - Apply compression stockings every day before rising.
 - Replace compression stockings every 3 months.
 - Stop smoking.
 - Maintain appropriate weight.
 - Walk briskly for 30 minutes twice each day.
 - Do not cross legs or stand for prolonged periods.
 - Elevate legs above heart.
 - Exercise ankles and calves often throughout day.
 - If nonambulatory, sit in rocking chair and use feet to propel vigorous rocking several times a day.
 - Avoid trauma to lower legs.
 - Do not wear high-heeled shoes.
 - Do not go barefoot.
 - Lower extremity neuropathic disease
 - Wear socks and hose.
 - Wear leather or suede shoes that are professionally fitted.
 - Wear new shoes for no more than 2 hours, and check feet for pressure areas after wearing them.

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6/25/24, 7:03 PM

SR - Wounds Due to Surgery, Injury, or Pressure

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Footnotes

[A] Wet to dry dressings are no longer recommended.(17)(21) [A in Context Link 1]

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12/12